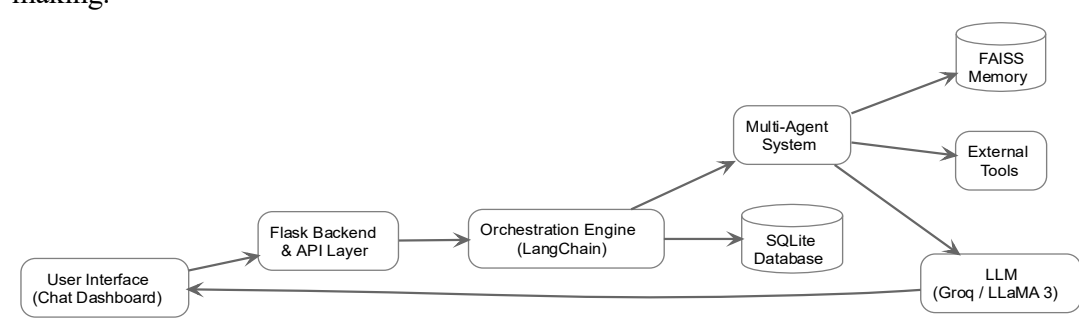


Introduction:

The Agent-Orchestration Framework with LangChain is an intelligent AI system developed to improve the processing of complex user queries through coordinated multi-agent execution. The framework combines Large Language Models (LLMs), LangChain orchestration, Retrieval-Augmented Generation (RAG), vector databases, persistent memory, and external tool integration within a unified architecture. Traditional AI assistants often rely on a single execution path, which can limit their ability to perform specialized reasoning, document analysis, tool usage, and content generation. The proposed framework overcomes these limitations by dynamically routing user requests to the most suitable execution pipeline.

System Architecture:

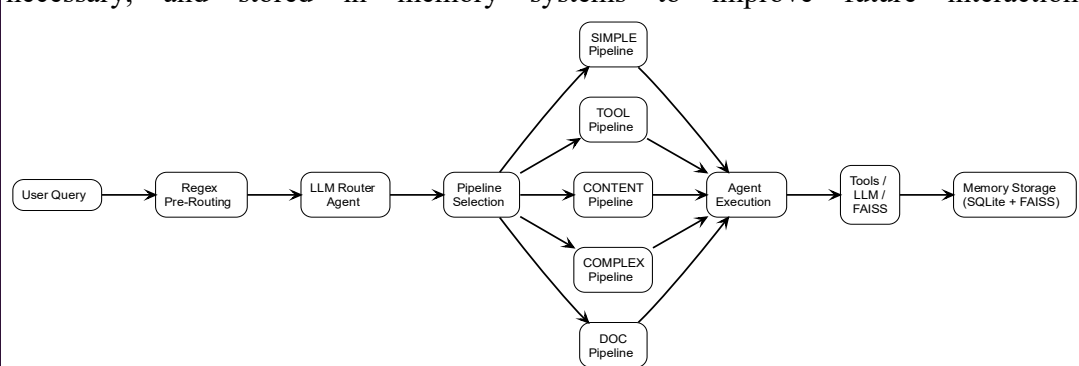
The system receives user queries through a web-based interface and forwards them to the orchestration engine. Queries are classified and routed to the appropriate pipeline for execution. Multiple agents collaborate to perform reasoning, tool execution, content generation, and document analysis. It utilizes LangChain, FAISS, SQLite, and Large Language Models to generate context-aware responses and support intelligent decision-making.



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graph LR; UI[User Interface Chat Dashboard] --> FB[Flask Backend & API Layer]; FB --> OE[Orchestration Engine LangChain]; OE --> MAS[Multi-Agent System]; MAS --> FM[(FAISS Memory)]; MAS --> ET[External Tools]; MAS --> LLM[LLM Groq / LLaMA 3]; MAS --> SD[(SQLite Database)]; SD --> LLM; LLM --> UI
```

Methodology:

The methodology begins with receiving a user query and performing route classification. Depending on the query type, the system activates the SIMPLE, TOOL, CONTENT, COMPLEX, or DOC pipeline. Specialized agents execute the required tasks, retrieve relevant information, and generate responses. The output is evaluated, refined when necessary, and stored in memory systems to improve future interactions.



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graph LR; UQ[User Query] --> RPR[Regex Pre-Routing]; RPR --> LRA[LLM Router Agent]; LRA --> PS[Pipeline Selection]; PS --> SP[SIMPLE Pipeline]; PS --> TP[TOOL Pipeline]; PS --> CP[CONTENT Pipeline]; PS --> COMPLEX[COMPLEX Pipeline]; PS --> DP[DOC Pipeline]; SP --> AE[Agent Execution]; TP --> AE; CP --> AE; COMPLEX --> AE; DP --> AE; AE --> TLLM[Tools / LLM / FAISS]; TLLM --> MS[(Memory Storage SQLite + FAISS)];
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Conclusion:	Maps to:	Name of Students	Name of Guide
	PO-1 to 5 and PO-9 to 12. PSO-1, PSO-2	Mr. Sidram Patil	Ms. M. H. Khan